

COMICCRAFT

DOCUMENT 4: SOLUTION ARCHITECTURE

PRESENTATION / CLIENT LAYER

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■ Web Browser (HTML5/CSS3/JavaScript + Jinja2 Templates) ■
■ ■■■ index.html (Input Form Page) ■
■ ■■■ comic_preview.html (Panel Viewer Page) ■
■ ■■■ export_success.html (Download Confirmation Page) ■
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■ HTTP Requests (POST/GET)
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API GATEWAY / SERVER

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■ FastAPI + Uvicorn Web Server ■
■ ■■■ CORS Middleware ■
■ ■■■ Static File Mount (/static/) ■
■ ■■■ GET / → Renders index.html ■
■ ■■■ POST /generate → Full pipeline (story + images) ■
■ ■■■ POST /generate-comic/json → JSON-only API endpoint ■
■ ■■■ GET /download/{id} → Serves exported PDF file ■
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■ Gemini ■ ■ Gemini ■ ■ Image Generation ■
■ Flash 1.5 ■ ■ Pro 1.5 ■ ■ Pipeline ■
■ (Outline) ■ ■ (Story) ■ ■ Stable Diffusion ■
■ Generates ■ ■ Expands ■ ■ / FLUX.1-schnell ■
■ 5-panel ■ ■ with rich ■ ■ Creates panel ■
■ JSON plan ■ ■ narration ■ ■ illustrations ■
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■ LAYOUT BUILDER ■
■ Assembles panel images with titles, narration captions, ■
■ dialogue balloons, and scene descriptions into layout ■
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■ PDF EXPORTER (FPDF2) ■
■ Compiles complete comic into multi-page PDF with ■
■ Unicode font support (DejaVu Sans), title page, ■
■ 5 panel pages, proper formatting ■
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DATA / STORAGE LAYER

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■ ■ ■ ■ ■ static/panels/ (Temporary generated PNG images) ■
■ ■ ■ ■ ■ static/exports/ (Generated PDF files for download) ■
■ ■ ■ ■ ■ .env (API key configuration - gitignored)■

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Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	User-facing web pages: input form for story parameters, interactive preview showing 5 panels with navigation, export success page with download link. Dark glassmorphism theme	HTML5, CSS3, Vanilla JavaScript, Jinja2
API Gateway / Server	FastAPI server handling all HTTP routing, input validation via Pydantic schemas, static file serving, CORS configuration, and response rendering (HTML/JSON)	FastAPI, Uvicorn, Pydantic
Gemini Flash Service	Generates structured 5-panel comic outline from user prompt - returns JSON with panel titles, scene descriptions, and key visual elements for each panel	Google Gemini 1.5 Flash API, gemini_flash.py
Gemini Pro Service	Expands the Flash outline into rich narrative content - full narration captions, character dialogue, and atmospheric scene descriptions for each panel	Google Gemini 1.5 Pro API, gemini_pro.py
Image Generation	Creates one AI-generated illustration per panel based on visual prompts derived from scene descriptions and selected art style	Hugging Face Diffusers, Stable Diffusion v1.5 / FLUX.1-schnell, image_generator.py
Layout Builder	Assembles panel images with overlaid text (titles, narration, dialogue) into structured comic panel layout with proper positioning	Python, Pillow (PIL), layout_builder.py
PDF Exporter	Compiles complete comic into downloadable multi-page PDF document with Unicode text support, proper margins, and formatting	FPDF2 (fpdf2), DejaVu Sans font, exporters.py